

SWAMINARAYAN UNIVERSITY



सर्व विद्या अधिपतये

**FACULTY OF ENGINEERING
DIPLOMA ENGINEERING**

SEMESTER-3rd

SUBJECT : BASICS OF OPERATING SYSTEM (DE104304)

UNIT 1: Introduction to Operating System

1. Explain the fundamental goals and services of an Operating System with a neat diagram.
2. Compare Multiprogramming, Time Sharing, Real-Time, Multithreading, and Distributed Operating Systems.
3. Explain the architecture and features of the Linux Operating System.
4. Describe the generations of Operating Systems with suitable examples.
5. Compare Linux and the latest Windows Operating System based on architecture, features, security, and applications.

UNIT 2: Process Management and Inter Process Communication

6. Explain the Process Life Cycle with a neat diagram. Describe the role of the Process Control Block (PCB).
7. Explain FCFS, Shortest Job First (SJF), and Round Robin Scheduling Algorithms with suitable examples.
8. Discuss CPU Scheduling Criteria (CPU Utilization, Throughput, Turnaround Time, Waiting Time, and Response Time).
9. Explain Inter Process Communication (IPC), Critical Section Problem, Semaphore, and Race Condition with examples.
10. Explain Deadlock. Discuss its characteristics, prevention, avoidance, detection, and recovery techniques.

UNIT 3: Memory Management

11. Explain Logical Address and Physical Address. Describe memory relocation and protection mechanisms with a neat diagram.
12. Explain Contiguous Memory Allocation. Differentiate between Fixed and Variable Partitioning.
13. Explain Internal Fragmentation, External Fragmentation, and Compaction with suitable examples.

14. Explain First Fit, Best Fit, and Worst Fit Memory Allocation techniques with examples.
15. Explain Paging and Segmentation with neat diagrams. Compare Paging and Segmentation.

UNIT 4: File Management System

16. Explain the File System. Describe file attributes, file operations, and file types.
17. Explain different Directory Structures with neat diagrams.
18. Explain File Allocation Methods (Contiguous and Linked Allocation) with their advantages and disadvantages.
19. Explain Disk Structure and SCAN & C-SCAN Disk Scheduling Algorithms with suitable examples.
20. Discuss File Protection mechanisms and Directory Protection in an Operating System.

UNIT 5: Linux and Shell Programming

21. Explain the basic architecture of Unix/Linux with a neat diagram.
22. Explain the purpose and syntax of Linux commands (pwd, cd, mkdir, rmdir, ls, cat, cp, rm, mv, wc, grep, sort, head, tail) with examples.
23. Explain text editors in Linux (vi, vim, and gedit) and their features.
24. Write and explain Shell Programs using command-line arguments, logical operators, and the test command.
25. Write Shell Scripts to demonstrate the use of if, case, for, and while statements with suitable examples.

*******DO YOUR BEST*******